

Flat Roofs, Flashings and Getting Water Away

Where a roof stops shedding and starts sealing — and every component that carries water off the building.

A flat roof cannot shed water by slope, so it relies on a continuous sealed surface. That changes where it fails: not at the fasteners but at the joins.

Single-ply membrane and roll roofing	Fail at loose joints and open seams resulting in water intrusion. On a sheet covering, always inspect the seams first.
Polyurethane foam roofing	A spray-applied closed-cell foam. Closed cells mean the bubbles are sealed, so the foam does not absorb water.
Scuppers	The rectangular openings in a parapet wall that let a flat roof drain through it. A roof enclosed by parapets has nowhere else for water to go.

YOU HAVE MISSED THIS 3 TIMES

Roofs surrounded by parapet walls may drain water through rectangular openings in the wall called: **Scuppers.**

YOU HAVE MISSED THIS 2 TIMES

Polyurethane foam roofing is a spray-applied _____ foam.
Closed-cell.

Flashings and penetrations

Every hole in a roof is a leak waiting to happen, so every hole gets a flashing. The exam asks about the ones with names and the one distance.

Roof jacks	Exhaust system flashings, made from galvanized steel, plastic or aluminum.
Washer on the head end	What fasteners in a lapped seam metal panel roof need. The washer seals the hole the screw just made in the middle of a water surface.

YOU HAVE MISSED THIS 4 TIMES

Fasteners with lapped seam metal panel roofs need a _____ on the head end.
Washer.

YOU HAVE MISSED THIS 2 TIMES

Exhaust system flashings are sometimes called:
Roof jacks.

The numbers here

Valley flashing must not be penetrated within this distance of the valley center	18 in
Chimney above the roof penetration	3 ft
Overflow drain above the roof drain	2 in

YOU HAVE MISSED THIS 3 TIMES

Valley flashing should not be penetrated by plumbing vents, appliance vents and other penetrations within ____ inches of the center of the valley.

Eighteen.

A valley is where two roof planes meet and concentrate their water into one channel — the fastest, deepest water on the whole roof. Eighteen inches either side of the center keeps penetrations out of that stream.

Off the roof and away from the house

Water leaving the gutter still has to get clear of the foundation, which is where the roof hands over to the site.

YOU HAVE MISSED THIS 2 TIMES

Downspouts should discharge onto a plastic or concrete tray called a:

Splash block.

YOU HAVE MISSED THIS 2 TIMES

Water from downspouts should not be:

Allowed to accumulate around the foundation.

A splash block spreads the discharge and stops it eroding a hole beside the foundation. The overflow drain sits two inches above the main roof drain so that a blocked primary drain announces itself by running out of the overflow rather than by ponding.

How long gutters last

The numbers here

Copper gutters	40 years
Aluminum gutters	20 to 30 years
Galvanized steel gutters	20 years

YOU HAVE MISSED THIS 2 TIMES

The typical life expectancy of copper gutters is:

40 years.

Copper doubles the life of the others, which is the only comparison worth holding: copper 40, everything else about 20.

Say it back

Cover the page. Answer each one out loud.

1. Where do single-ply and roll roofs fail?
2. What kind of foam is sprayed polyurethane roofing, and why does it matter?
3. What drains a roof enclosed by parapet walls?
4. What are roof jacks?
5. What do lapped seam metal panel fasteners need, and what does it do?
6. How far from the valley center must penetrations stay, and why?
7. What does a downspout discharge onto, and what must the water not do?
8. Life expectancy of copper, aluminum and galvanized gutters?
9. How far above the roof drain does the overflow sit, and what does that achieve?